

WEEK 1

THE AI SHIFT

AI, Business & Competitive Transformation
Graduate • 3-hour session • Thursday, August 27,
2026

01

MIS 7397 • WEEK 1



COLD OPEN

You already used AI today. Where?

**PHONE**

Face unlock • keyboard • camera

**COMMUTE**

Maps • traffic • rideshare

**INBOX**

Spam • priority • suggested replies

**SHOPPING**

Search • ranking • recommendations

**MEDIA**

Feeds • playlists • discovery

**WORK**

Copilots • forecasting • fraud

QUICK POLL

How many AI-assisted decisions have touched your day before this class?

2-0

5-3

10-6

+ 10

The point: AI is already embedded in business processes—even when the user never sees a robot.

COURSE VISION

The promise of this course

**By December, you should be able to sit between the business
and the technical team—and make a better AI decision**

**IDENTIFY**

High-value AI
opportunities

**COMMUNICATE**

With technical and
executive teams

**EVALUATE**

Value, evidence,
limitations & risk

**LEAD**

Implementation and
responsible adoption

NO PRIOR PROGRAMMING REQUIRED

**UNDERSTAND****IDENTIFY****LEARN****BUILD****EVALUATE****LEAD**

MIS 7397: Introduction to Artificial Intelligence for Business

What students will learn and what makes this course distinctive



COURSE VISION

A practical, business-focused AI course that helps students understand, evaluate, and lead AI initiatives without requiring a computer science background.

WHAT WE WILL TEACH

1



AI FOUNDATIONS

- AI concepts
- business problem framing
- data readiness



2



CORE AI METHODS

- machine learning
- deep learning
- NLP
- computer vision



3



MODERN AI

- generative AI
- LLMs
- prompting
- RAG
- agents



4



BUSINESS IMPLEMENTATION

- no-code labs
- prototypes
- ROI
- build vs. buy
- security



5



STRATEGY & RESPONSIBLE AI

- governance
- ethics
- privacy
- bias
- organizational change



STUDENT OUTCOME

Students leave able to identify high-value AI use cases, communicate with technical teams, evaluate AI opportunities and risks, and present an executive-level implementation recommendation.



DISTINGUISHED FEATURES



Designed for business students



Hands-on no-code / low-code activities



Covers current AI topics beyond the textbook



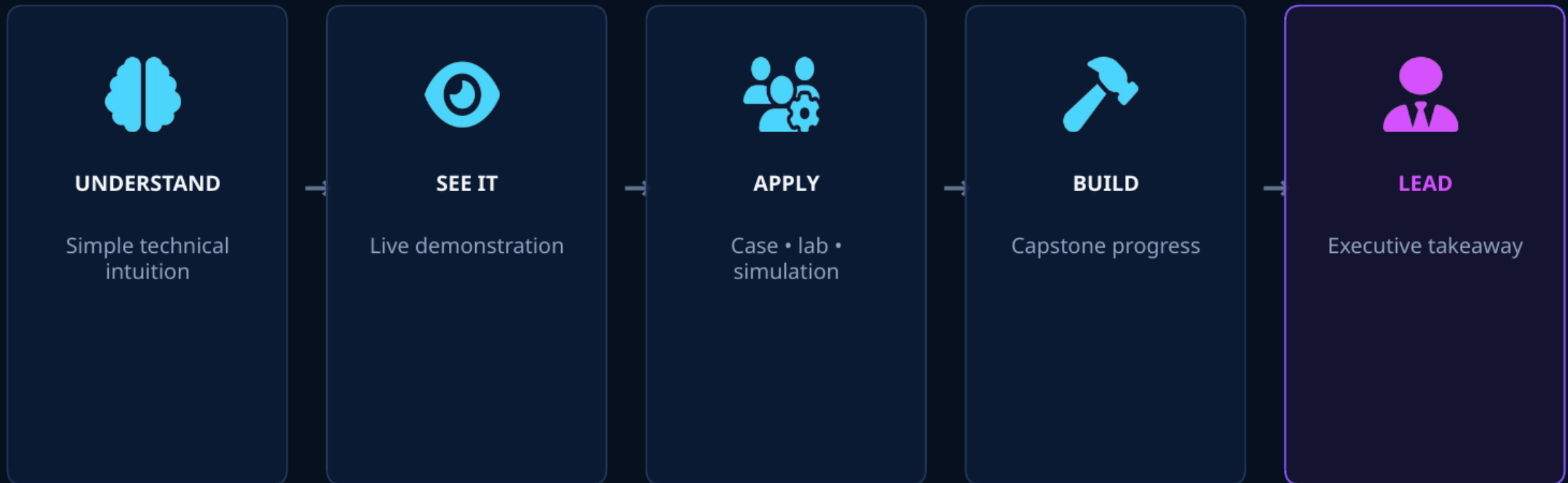
Semester-long business capstone



Focus on real business value, risk, and executive decision-making

HOW WE LEARN

Every class follows the same learning rhythm

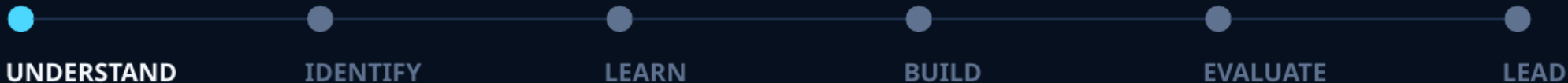


“.The objective is not “know the vocabulary.” It is “make a defensible decision

WEEK 1 OUTCOMES

Today: build the mental model

- 01** — **Distinguish** AI, automation, analytics, ML, deep learning, and GenAI.
- 02** — **Recognize** Six common business problems that AI can address.
- 03** — **Connect** AI capability to a decision, workflow, KPI, and business value.
- 04** — **Challenge** AI claims by asking about evidence, risk, and human oversight.
- 05** — **Launch** Your semester-long AI Opportunity-to-Deployment capstone.



CURRENT AI BRIEFING

AI NOW • AUGUST 27, 2026

Three signals to watch: capability, adoption, and governance

**CAPABILITY**

Aug. 25

Claude adds persistent memory

Memory now works across chat + Cowork. Users can edit or delete remembered topics; sensitive topics are excluded unless enabled.

BUSINESS SIGNAL

AI assistants are becoming more persistent and context-aware—not just one-session chatbots.

Anthropic Release Notes • Aug. 25, 2026

**ADOPTION**

This week

Cisco deploys agents to 90,000 employees

Cisco rolled out a personal AI agent, “MyAgent,” to its global workforce—an unusually broad enterprise deployment.

BUSINESS SIGNAL

Agentic AI is moving from isolated pilots into everyday employee workflows and operating infrastructure.

The Wall Street Journal • Aug. 27, 2026

**GOVERNANCE**

Aug. 2

EU AI Act enforcement is now active

The EU AI Act became broadly applicable this month. Transparency rules are in effect, with the AI Office and national authorities responsible for enforcement.

BUSINESS SIGNAL

AI governance is becoming an operating requirement: inventory, transparency, accountability, and controls.

European Commission • effective Aug. 2, 2026

THE PATTERN**PERSISTENT** AI remembers context**EMBEDDED** AI enters workflows**GOVERNED** controls become mandatory

CLASS QUESTION: Which signal will change business fastest—and which creates the biggest management challenge?

WHAT IS AI?

Artificial intelligence is a capability—not a product

AI systems perform tasks that normally require aspects of human intelligence.



PERCEIVE



LEARN



REASON



GENERATE



DECIDE / ACT

A better manager asks:

- ✓ What task is being performed?
- ✓ What evidence/data does it use?
- ✓ What decision or workflow changes?
- ✓ What happens when it is wrong?

.The label “AI” matters less than the capability and consequences

DO NOT LET THE BUZZWORD WIN

AI $\neq$ automation $\neq$ analytics



AUTOMATION

Follows explicit instructions

"If invoice > \$10K, route to manager."

RULES



ANALYTICS

Describes or explains data

"Revenue fell 8% in the Southwest."

INSIGHT



AI

Infers patterns or generates output

"This customer has 82% churn risk."

INFERENCE

.Real systems often combine all three

ONE PICTURE TO REMEMBER

The vocabulary map: AI → ML → Deep Learning → GenAI

ARTIFICIAL INTELLIGENCE

Broad field: systems that perform intelligent tasks

MACHINE LEARNING

Systems learn patterns from data rather than only explicit rules

DEEP LEARNING

Multi-layer neural networks learn complex representations

GENERATIVE AI

Creates new text, images, audio, code, and more

THREE AI MODES SHAPING BUSINESS

Predict → Create → Act

01



PREDICTIVE AI

What is likely to happen?

Churn • fraud • demand • risk

02



GENERATIVE AI

What should we create?

Text • image • code • summaries

03



AGENTIC AI

What action should we take next?

Retrieve • use tools • execute workflows

.These are complementary—not a timeline where one replaces the others

CAPABILITY LENS

Six business problems AI is especially good at

**CLASSIFY****What category?**

spam • intent • defect

**PREDICT****What is likely?**

demand • churn • risk

**RECOMMEND****What next/best?**

product • content • action

**GENERATE****Create something**

text • image • code

**OPTIMIZE****Best allocation?**

routes • schedules • pricing

**DETECT****What is unusual?**

fraud • anomaly • failure

WEEK 1 CHECKPOINT

AI Myth vs. Reality

MYTH AI = ChatGPT



ChatGPT is one AI product; AI includes prediction, vision, recommendations, optimization, robotics, and more.

MYTH Fluent = correct



A confident, polished answer can still be wrong. Output quality must be evaluated.

MYTH AI is objective



Models can inherit bias from data, labels, objectives, and deployment choices.

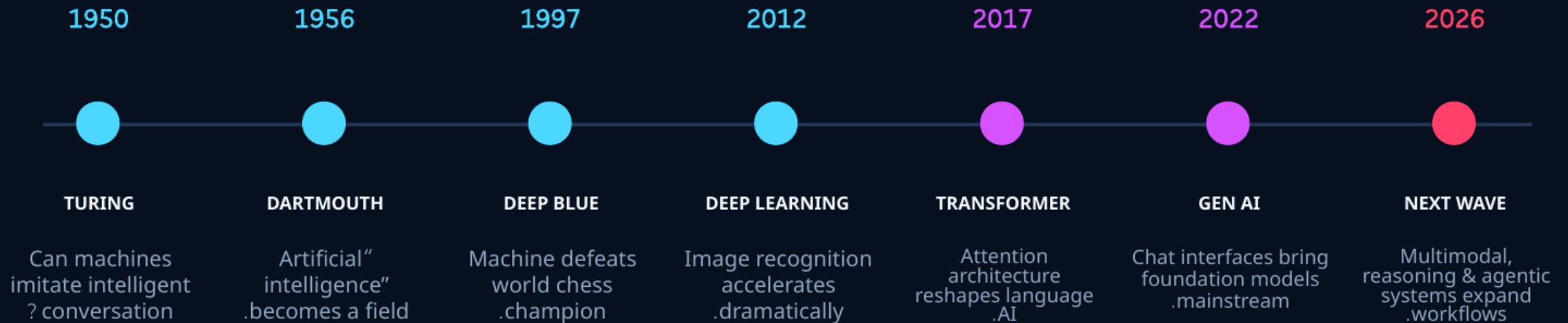
MYTH More automation = better



Automation can amplify a bad process or scale a small error into a large problem.

HOW WE GOT HERE

A 70-year journey in 90 seconds



.Progress has come in waves—not a straight line

THE ACCELERATION

Why now? Four forces converged

**DATA**

Digitized interactions and
enterprise data

+

**COMPUTE**

GPUs • cloud • specialized
hardware

+

**ALGORITHMS**

Deep learning •
transformers • foundation
models

+

**INTERFACES**

APIs • chat • copilots • no-
code tools

.Technology became easier to access at the same time that capability improved

**UNDERSTAND****IDENTIFY****LEARN****BUILD****EVALUATE****LEAD**

PART II

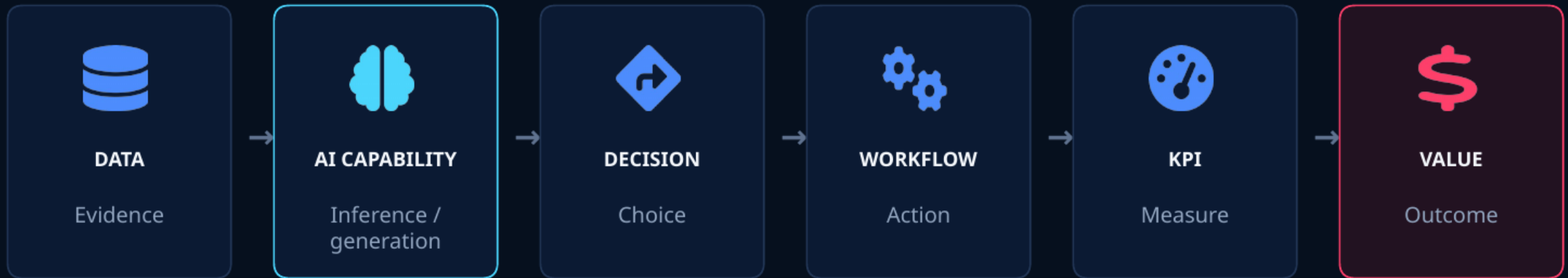
FROM AI CAPABILITY TO BUSINESS VALUE

An AI model creates no value by itself. Value appears when it changes a decision or workflow.



FROM MODEL TO OUTCOME

The business-value chain



Example: purchase history → churn prediction → retention offer → agent outreach → retained customers → recurring revenue

NO KPI = NO BUSINESS CASE

WHERE VALUE SHOWS UP

Five patterns of AI business value

**AUTOMATE**

**Reduce
repetitive effort**

speed • cost

**AUGMENT**

**Improve human
decisions**

quality • capacity

**PERSONALIZE**

Tailor at scale

conversion • loyalty

**DISCOVER**

**Find patterns
humans miss**

risk • insight

**CREATE**

**Generate new
content/work**

cycle time •
experimentation

.A strong use case usually names the value pattern and the metric

BUSINESS USE CASES

AI is not a “tech department” topic

**MARKETING**

recommend • personalize • generate

**FINANCE**

forecast • detect fraud • explain

**OPERATIONS**

optimize • predict failure • schedule

**HR**

match skills • summarize • support

**CUSTOMER**

classify • retrieve • respond

**KNOWLEDGE**

search • summarize • draft

.The differentiator is rarely “having AI.” It is redesigning the workflow around it better than competitors

SEE IT

LIVE DEMO: one customer problem, three AI approaches

CUSTOMER MESSAGE

"I was charged twice and support closed my ticket. If this is not fixed today, I am leaving."

Business goal: resolve quickly while reducing churn and policy risk.



RULES

Keyword "charged" → Billing queue



PREDICT

Churn risk = 0.86 → Priority = High



GENERATE

Draft empathetic response + next steps

? DEMO QUESTION: What should the AI decide—and what should a human still control

DEMO DEBRIEF

Same problem. Different capability. Different risk.

	BEHAVIOR	STRENGTH	WATCH OUT
RULES	Deterministic	Easy to audit	Brittle
PREDICTIVE ML	Probabilistic	Scales decisions	Needs data + monitoring
GENERATIVE AI	Flexible	Handles language	Can hallucinate / vary

.Management question: choose the least complex system that produces acceptable value and risk

LIMITS EVERY MANAGER SHOULD
ANTICIPATE

Powerful does not mean reliable



HALLUCINATION

Plausible but false output



BIAS

Unequal outcomes



DRIFT

Performance changes over time



PRIVACY

Sensitive data exposure



SECURITY

Manipulation / prompt injection



COST

Compute, latency, operations

RESPONSIBLE AI PREVIEW

Capability $\neq$ permission

? CAN WE**? SHOULD WE****? WHO IS ACCOUNTABLE**

Technical feasibility

Business / human consequences

Governance + oversight

.We will revisit this question in every major topic

MINUTE BREAK • BE BACK IN 10-10

APPLY

ACTIVITY: Is this actually AI?



TEAMS OF 3-4

For each scenario,
classify it as:

- Not AI / simple rule
- Analytics
- Predictive ML
- Generative AI
- Could be multiple

Then answer TWO management questions

1 What evidence would convince you it works?

2 What happens when it is wrong?

minutes team work • 7 minutes debrief 8



UNDERSTAND

IDENTIFY

LEARN

BUILD

EVALUATE

LEAD

TEAM WORK

Six scenarios. Six classifications.

A

An Excel formula calculates a 7.25% sales tax.

B

A system routes invoices over \$10,000 to a manager.

C

A dashboard shows last quarter's revenue by region.

D

A spam filter learned from thousands of labeled emails.

E

A retailer recommends products from browsing + purchase behavior.

F

A model drafts a personalized response to a customer complaint.

? Bonus: Which scenario is most likely to be called "AI" by marketing even if it is not

DECISION 01

YOU ARE THE EXECUTIVE

A national retailer proposes an AI refund agent.

**95%**

correct policy decisions

**−48%**

average handling time

**\$1.8M**

estimated annual savings

**100K**

refund cases per year

YOUR DECISION

**DEPLOY****PILOT****REJECT**

.Vote first. Explain second

SCALE CHANGES THE RISK

The twist: 5% error × 100,000 decisions

5,000

potentially incorrect refund decisions /
year

Some errors involve high-value items
and policy exceptions

NOW WHAT?



**Human
review**

Route low-confidence / high-value cases



Thresholds

Automate only within safe limits



Monitoring

Track error type, not only accuracy



Fallback

Allow fast appeal / reversal

Executive lesson: model performance + error consequence + control design = deployment decision

SEMESTER CAPSTONE

FROM AI IDEA → EXECUTIVE RECOMMENDATION

You will not write a paper about AI. You will develop, test, govern, and pitch a realistic AI-enabled business initiative.



CAPSTONE JOURNEY

One project. Six checkpoints.

OPPORTUNITY

DATA + FEASIBILITY

PROTOTYPE

EVALUATE

GOVERN

PITCH

01

02

03

04

05

06

Problem • users •
baseline • KPIData • constraints • non-
AI alternativeNo-code/low-code AI
demonstrationMetrics • testing • ROI •
human reviewRisk • privacy • security •
monitoringRoadmap • economics •
executive
recommendation

The question gets harder as the semester progresses: "Could this work?" → "Should this
"? organization actually do it



UNDERSTAND

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EXECUTIVE TAKEAWAY

Leave Week 1 with three ideas

1 AI is not one thing.

Choose the capability that fits the task—not the trend.

2 AI value lives in the workflow.

Connect model → decision → KPI → business outcome.

3 Performance is not enough.

Ask what happens when the system is wrong—and who controls it.

BEFORE NEXT CLASS

MindTap Ch. 1 selected work + AI
Opportunity Observation

EXIT TICKET

Name one AI use case + KPI + one failure
you would monitor.

NEXT WEEK → FROM BUSINESS PROBLEM TO AI OPPORTUNITY

MACRO LENS • GENERATIVE AI • 5 MIN

\$100M

I

**AND IT'S STILL
GUESSING.**